

Compare misspecification of covariates, functional forms and missing covariates.

We could use the variation measure (e.g. derive residuals for explained variation compared to true variation by confounders), to consider how much mis-specification. How robust the

Paper	Classification	Selection of features
Clarke et al. (2018) - Causal models adjusting for time-varying confounding—a systematic review of the literature (1)	TMLE	[Method already covered]
Talbot et al. (2025) - Guidelines and Best Practices for the Use of Targeted Maximum Likelihood and Machine Learning When Estimating Causal Effects of Exposures on Time-To-Event Outcomes (2)	TMLE	[Method already covered]
Gabriel et al. (2023) - Propensity weighting plus adjustment in proportional hazards model is not doubly robust (3)	IPW + Cox PH regression + Standardisation	• Demonstrates that combining IPW weights with a weighted Cox model is not doubly robust, as consistency requires both models to be correct. In addition, we need to have a correct model for censoring. Hence, we also need IPCW with IPW, and sufficient control for confounders to satisfy conditional independent censoring assumption in the outcome model • This applies only to PH models. This makes sense since PH cannot hold in the conditional and the marginal model. Worth considering both PH and VPH models. In addition this is where additive hazards models might be preferred in causal inference due to collinearity, causal interpretability and including time-dependent effects • Clarifies that this approach does not correspond to the AIPW influence function and therefore cannot recover marginal causal effects • Should avoid log hazard ratios as estimates for causal effects

Vin Hage et al. (2025) - Doubly Robust Estimation of Marginal Cumulative Incidence Curves for Competing Risk Analysis (4)	AIPW	[Method already covered]
Frank et al. (2011) - Doubly Robust Estimation of Causal Effects (5)	AIPW	[Method already covered]
Kurz et al. (2022) - Augmented Inverse Probability Weighting and the Double Robustness Property (6)	AIPW	[Method already covered]
Syrjämä et al. (2021) - Inverse probability weighting and doubly robust standardization in the relative survival framework (7)	AIPW	[Method already covered]
Deane et al. (2023) - A comparison of different methods to adjust survival curves for confounders (8)	AIPW	[Method already covered]
Choi et al. (2022) - Doubly-robust methods for differences in restricted mean lifetime using pseudo-observations (9)	AIPW	[Method already covered]
Meibohm et al. (2025) - Adjusting for time-varying treatment effects in randomized clinical trials: the danger of extrapolation and how to avoid it (10)	AIPW (tv-based / SSMSTMA variant)	• Focuses on estimation of difference in restricted mean lifetime, which may be of relevance in presence of non-proportional HR • Aimed at targeting the hypothetical standard (ICHE) in presence of time-varying treatment switches and right-censored survival times • Rules on randomization as instrumental variable • R software provided • Provides closed form formula • Assumes no baseline confounding (i.e. treatment is randomized) • Provides SAS code
Kawahara et al. (2020) - Doubly robust estimator of risk in the presence of censoring dependent on time-varying covariates: application to a primary prevention trial for coronary events with prevascular (11)	AIPW	[Method already covered]
Smilde et al. (2022) - Introduction to computational causal inference using reproducible Stata, R, and Python code: A tutorial (12)	AIPW	[Method already covered]
Wang et al. (2023) - Multiply robust estimator for the difference in survival functions using pseudo-observations (13)	AIPW	• Combines multiple propensity score (PS) models and multiple outcome regression (OR) models • Targets the difference in survival function • Focuses on baseline confounding only
Chariton et al. (2022) - G-computation and doubly robust standardisation for continuous-time data: A comparison with inverse probability weighting (14)	DRS	[Method already covered]
Vassterlandt (2019) - On regression adjustment for the propensity score (15)	DRS	[Method already covered]
Saari et al. (2020) - The Calculus of M-Estimation in $\mathbb{R}$ with ψ-genψ (16)	M-estimation	[Method already covered]

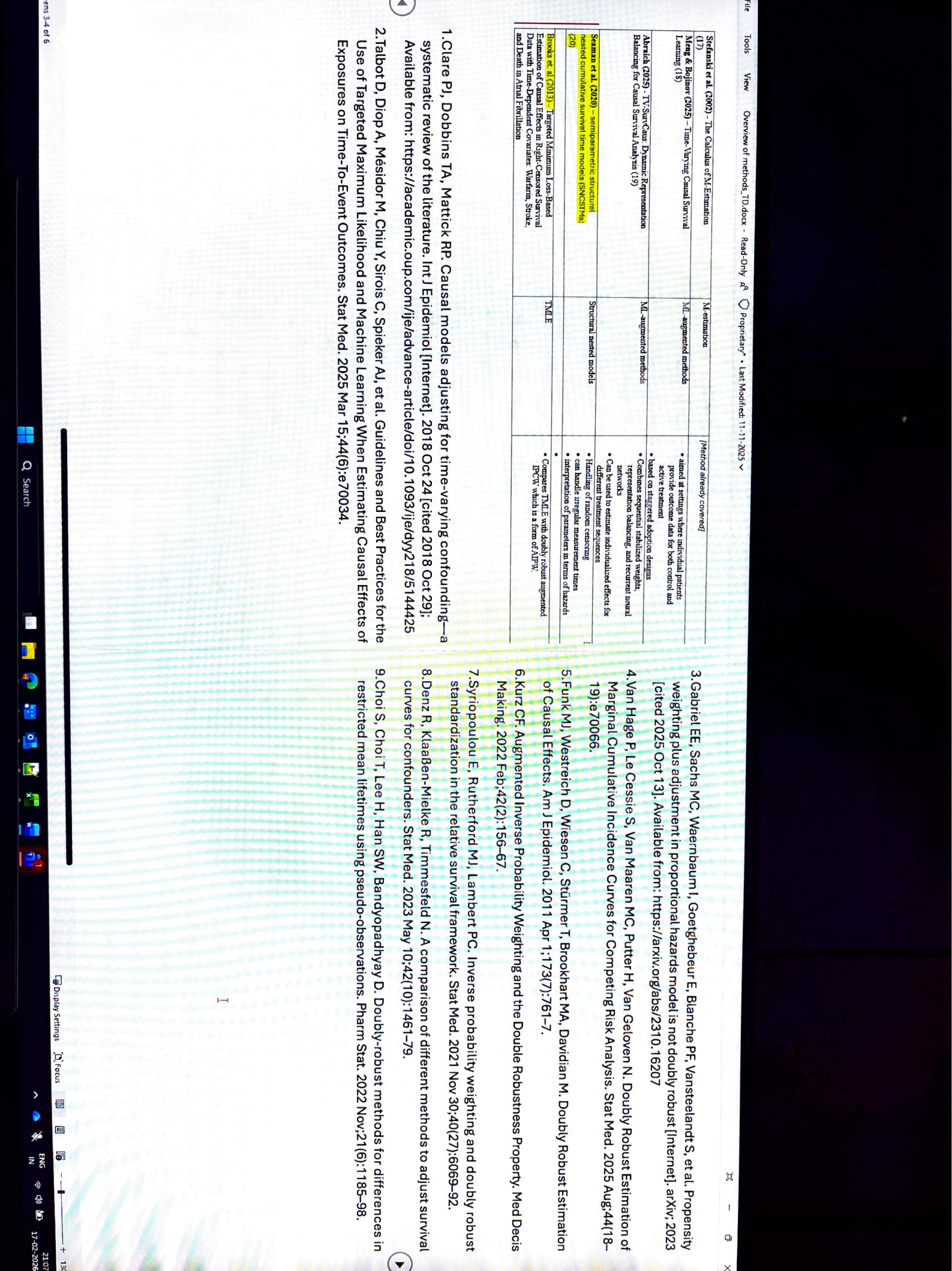
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	Method already covered
Stefanski et al. (2002) - The Calculus of N: Estimation (17)	ML augmented methods
Meng & Böhmer (2023) - Time-Varying Causal Survival Learning (18)	• aimed at settings where individual patients provide outcome data for both control and active treatment • based on staggered adoption designs
Abraha (2023) - TV-SurCaus: Dynamic Representation Balancing for Causal Survival Analysis (19)	• Combines sequential, stabilized weights, representation balancing, and recurrent neural networks • Can be used to estimate individualized effects for different treatment sequences • Handling of random censoring • can handle irregular measurement times • interpretation of parameters in terms of hazards
Seaman et al. (2020) - semiparametric structural nested cumulative survival time models (SMCS1746) (20)	• Compare TMLE with doubly robust augmented IPCW, which is a form of AIPW.
Brooks et al. (2013) - Targeted Minimum Loss-Based Estimation of Causal Effects in Right-Censored Survival Data with Time-Dependent Covariates: Warfarin, Stroke, and Death in Atrial Fibrillation	TMLE

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